

Produit : Electronic / Measuring devices

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INDIA

Product's definition

Category

In which category the product is classified ?

Different products falling under the categories of Lamps are defined under different standards -

IS 10322 (Part 5 / Sec. 6): 2013 - Luminaries (Part 5: Particular requirements, Section 6: Handlamps)

IS 302-1:2008 - Safety of Household and Similar Electrical Appliances

Other standards may include (subject to Decathlon products falling under this category) -

IS 16102(Part 1) - Self-Ballasted LED Lamps for General Lighting Services Part 1 Safety Requirements

IS 418 - Tungsten filament general service electric lamps (upto100 W)

Definition

What is the exact definition of the product ?

And if it exists, what is the interpretation of the authorities regarding it ?

While there is not exact definition under law for Lamp, we can look at the description provided under the HSN Code -

HSN 8513 the term self-powered lamp covers - "Portable electric lamps designed to function by their own source of energy (e.g., dry battery, accumulator, or solar cell)."

Examples -

- Cap lamps (used in mining and industrial applications)
- Rainproof torches
- LED lanterns

- Flashlights
- Headlamps

[Comments](#)

[Comments](#)

Labelling

[Language](#)

[In which language must the information on the product or on the packaging be indicated? If English is sufficient please indicate NA.](#)

Note: Generally All the particulars and Labelling are made in English Language.

By The Legal Metrology (Packaged Commodities) Rules, 2011

Chapter II Provisions Applicable to Packages Intended for Retail Sale.

[Mandatory labelling on product](#)

[What are the mandatory labelling on the product ?](#)

[Mandatory labelling on packaging](#)

[What are the mandatory labelling on the packaging ?](#)

As per the Legal Metrology Act, 2009 & Packaged Commodities Rules, 2011 - Rule 6

Every packaged product should have the following declarations in the form of an MRP Label affixed on it -

1. Maximum Retail Price
2. Month and Year of Import/Manufacturer
3. Name and Address of the Importer/Manufacturer/Packer
4. Generic Name of the Product (Nature of Product)
5. Size of the product (ml, l, cm, m, gms, kg)
6. Quantity (number of units in the package)

7. Price Per Unit

8. Country of Origin

9. Best Before / Use By Date (in case of perishable products)

The above declarations is updated on the MRP tool developed for India. Suppliers across the world need to feed in the model code / item code and the sticker auto generates and can be printed. The printed sticker should be affixed on the package ensuring that no other declarations in English are covered, and there is no fold / restriction to the MRP sticker. It should be on a flat surface.

NOTE:

For Electronic Goods Manufactured/Imported after the 15th of July 2022 - the package of such product shall, for a period of one year from such date, declare the name of the manufacturer or packer or importer, as the case may be, on the package itself and such declaration shall also inform the consumers to scan the QR code for the address and other related information, in case such information is declared through the QR Code and not declared on the package itself.

Mandatory labelling on the instructions manual

What are the mandatory labelling on the instructions manual (if one is required) ?
In which format, do we have to deliver the instruction manual to our customers ?

Not Applicable

Warnings - General ones

Are general warnings required ? If yes, which ones ?

Not Applicable

Warnings - Specific ones

Are specific warnings required ? If yes which ones ?

Not Applicable

Target Age

Is there any target age applicable? If so, does it have to be indicated and in which form?

Not Applicable

Sanctions / Fines

What are the sanctions / fines in case of non conformity ?

By The Legal Metrology (Packaged Commodities) Rules, 2011

Chapter II Provisions Applicable to Packages Intended for Retail Sale:

20. Action to be taken on completion of inspection of packages at the premises of the manufacturer or the packer:- (1) If it appears from the report referred to in sub-rule (3) of rule 19 that,-

(a) the statistical average of the net quantity contained in the packages drawn as samples under that rule is lesser than the quantity declared on the packages or on the labels affixed thereto, or any such package shows an error in deficiency greater than the maximum permissible error, or

(b) any such package does not bear thereon or on label affixed thereto the declarations to be made under these rules, the Director, Controller or any Legal Metrology Officer shall take action as given below, namely:-

(i) seize the packages drawn by him as samples and shall take adequate steps for the safe custody of seized packages until they are produced in the appropriate court as evidence;

(ii) based on the evidence initiate action for violations of the provisions of the Act and these rules:

Provided that no such action shall be taken if fresh tests are carried out under sub-rule (4) of rule 19, but if after such fresh tests any error or omission as is referred to in this sub-rule is detected, the Director, Controller or any Legal Metrology Officer shall take appropriate action as specified in this sub-rule in accordance with the provisions of the Act against the manufacturer or, as the case may be, the packer.

(3) The disposal of the seized packages shall be made in accordance with the provisions of the Code of Criminal Procedure, 1973 (2 of 1974).

Chapter VII GENERAL

32. Penalty for contravention of Rules.-(1) Whoever contravenes the provisions of rules 27 to 31, he shall be punished with fine of four thousand rupees.

(2) Whoever contravenes any other provision of these rules, for the contravention of which no punishment has been provided either in the Act or in the rules, he shall be punished with fine of two thousand rupees.

[Comments](#)

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By The Legal Metrology (Packaged Commodities) Rules, 2011

Chapter II Provisions Applicable to Packages Intended for Retail Sale:

18. Provisions relating to wholesale dealer and retail dealers: (Refer to authority attached)

19. Inspection of quantity and error in packages at the premises of the manufacturer or packer:- (Refer to authority attached)

21. Inspection of quantity and error in packages at the premises of the wholesale dealer or retail dealer :- (Refer to authority attached)

23. Deceptive packages to be repacked or in default to be seized: (Refer to authority attached)

Chapter IV EXEMPTIONS

26. Exemption in respect of certain packages: - Nothing contained in these rules shall apply to any package containing a commodity if-

(a) the net weight or measurement of the commodity is ten gram or ten milliliter or less, if sold by weight or measurement.(**)

Chapter VI REGISTRATION OF MANUFACTURERS, PACKERS AND IMPORTERS

27. Registration of manufacturers, Packers and importers :(Refer to authority attached)

28. Registration of shorter address permissible :.. (Refer to authority attached)

29. Registration of manufacturers and packers, etc.: .. (Refer to authority attached)

Chapter VII GENERAL

31. (1) Any advertisement mentioning the retail ale price of the pre-packaged commodity shall contain a declaration as to the net quantity or number of the commodity contained in the package.

(2) The font size of the net quantity in the advertisement shall be same as that of retail sale price.

30. Compilation of lists of manufacturers or packers and their circulation...

33. Repeal and savings ...

Environment

Recycling - Eco taxes

Which regulations concerning ecodesign and recycling apply to this product? Are there any ecotaxes applicable to this category of product?

Not Applicable

Labelling

What are the mandatory labellings to put on the product or on the packaging regarding the environmental rules ?

Not Applicable

Technical rules / Standards

General safety requirements

What are the general safety requirements ? Does a risk assessment have to be conducted ? If yes, does it concern specific risk ?

IS 10322-5-5 (1987) : Luminaires, Part 5: Particular requirements, Section 5: Flood Light (Superseding IS:1947) [ETD 24: Illumination Engineering and Luminaries]

i.e Indian Standard – SPECIFICATION FOR LUMINAIRES -Part 5 Particular Requirements-Section 5 Flood-Lights

3. GENERAL TEST REQUIREMENTS

3.1 The provisions of 3 of Part 1 of this standard shall apply. The tests shall be carried out in the order listed in this standard.

13. TESTS

13.1 Classification of Tests

13.1.1 Type Tests – The following shall constitute the type tests:

- a) Visual examination (see 13.2),
- b) Mechanical strength test (see 13.3)
- c) Endurance tests and thermal tests (see 13.4),
- d) Resistance to dust and moisture (see 13.5)
- e) Insulation resistance and electric strength (see 13.6),
- f) Resistance to heat, fire and tracking (see 13.7), and
- g) Photometric test (see 13.8).

13.1.1.1 Criterion of acceptance – Samples for type testing shall be mutually agreed between the purchaser and the supplier. These samples shall be submitted for testing together with relevant data. The testing authority shall issue a type approval certificate if the luminaires comply with the requirements given in 13.1.1.

In a case of failure in any one type test, the testing authority may call for fresh samples not exceeding twice the number of original samples and subject them to test in which failure occurred. If in repeat tests, no failure occurs, the tests shall be considered to have been satisfied.

13.1.2 Acceptance Tests – The following shall constitute the acceptance tests;

- a) Visual examination (see 13.2),
- b) Resistance to dust and moisture (see 13.7),
- c) Insulation resistance and electric strength (see 13.7), and
- d) Photometric test (see 13.8).

13.1.2.1 The number of samples for acceptance shall be as agreed upon between the purchaser and the supplier. However, a recommended plan of sampling is given in Appendix A.

13.1.3 Routine Tests – The following shall constitute the routine tests:

a) Visual examination (see 13.2), and

b) Insulation resistance and electric strength (see 13.6).

13.2 Visual Examination – The luminaires shall be examined visually for external finish, workmanship and electrical connections.

13.3 Mechanical Strength Test – The provisions of 5 of Part 4 of this standard shall apply.

13.4 Endurance Test and Thermal Test – The provisions of 6 of Part 4 of this standard shall apply.

13.5 Resistance to Dust and Moisture – The provisions of 2 of Part 4 of this standard shall apply.

13.6 Insulation Resistance and Electric Strength Test – The provisions of 3 of Part 4 of this standard shall apply.

13.7 Resistance to Heat, Fire and Tracking – The provision of 7 of Part 4 of this standard shall apply.

13.8 Photometric Test – The photometric performance shall be determined by a suitable photometric tests as agreed between the purchaser and the supplier. A recommended photometric test for flood-lights fitted with tungsten filament lamps is given in Appendix B.

13.8.1 Flood-lights fitted with tungsten filament lamps shall meet the photometric requirements specified in 12.

13.8.2 The photometric performance shall be determined by the test method given in IS 13383 (Part 3): 1992 Photometry of luminaires – Method of measurement : Part 3 Luminaires for flood lighting.

APPENDIX A (for Clause 13.1.2.1) SAMPLING PLAN FOR ACCEPTANCE TESTS (Please refer to PDF file attached)

APPENDIX B (For Clause 13.8) PHOTOMETRIC TEST FOR FLOOD-LIGHTS (Please refer to PDF file attached)

IS 418:2004 i.e Indian Standard-TUNGSTEN FILAMENT LAMPS FOR DOMESTIC AND SIMILAR GENERAL LIGHTING PURPOSES (Fourth Revision)

12 SAMPLING

12.1 Principles of Sampling

The lamps for testing shall be selected so as to ensure proper representation.

NOTE — It should first be ascertained that the values of the rated luminous flux comply with the requirements of the relevant lamp data sheet.

It is not necessary to replace an accidentally broken lamp if the result of the test (approval or rejection) is not affected by its replacement, provided the required quantity of lamps for any subsequent test is available. If replaced, a broken lamp shall be neglected in calculating the test results.

NOTE — Accidentally broken lamps include, for example, lamps damaged during handling and transportation and also lamps becoming defective for reasons, which are not connected with the purpose of a particular test being applied.

For batch and comparability testing, some lamps additional to the test quantity shall be selected. These lamps shall only be substituted for lamps of the test quantity, if necessary to make up the required number of lamps for the test.

12.2 Sampling for Whole Production Testing (Applicable for Manufacturers)

12.2.1 Pre-compliance Testing for Certification Purposes

Sampling for the pre-compliance test is given in C-2.

12.2.2 Manufacturer's Test Data

The manufacturer shall make available all the data of his finished product tests so far as these relate to the lamp types on the manufacturer's nominated list and are pertinent to the requirements of this standard.

These data shall refer to a sufficient number of lamps, selected over a 12-month period, so as to be representative of the whole production. To meet this requirement there shall be provided:

Manufacturers' test data on:

— for the four largest groups (or all groups if there are less than four), at least 200, 300 and 200 lamps for, respectively, ITQ, RTQ and LTQ with a minimum of 40, 60 and 40 lamps per group for, respectively, ITQ, RTQ and LTQ. If the LTQ of 200 represents more than 0.01 percent of the production, then only 0.01 percent or 40 lamps, whichever is the greater, need be tested;

— for each of the other groups, which together with the four largest groups make up at least 75 percent of the production, a minimum of 20, 30 and 20 lamps for, respectively, ITQ, RTQ and LTQ;

— where a number of types make up a group, test quantities shall be selected from each of those types which make up at least 50 percent of the production for that group;

— for each type for which data has to be presented to meet the above requirements a minimum of 20, 30 and 20 lamps for, respectively, ITQ, RTQ and LTQ;

Providing the above requirements have been met, any types in the manufacturer's records for which there are less than the quantities for ITQ, RTQ and LTQ of 20, 30 and 20, respectively, shall not be considered.

All tests need not necessarily be carried out on the same lamps. The RTQ may contain other individuals than the ITQ, but the LTQ shall be carried out on individuals selected at random from lamps, which have passed the rating test.

NOTE — This is because the life test results are dependent on the rating test results, this not being the case for inspection test results.

As it may be difficult at the time of sampling to forecast the 12-month production of a type, as a fraction of the whole, percentage values in this section are to be regarded as guidelines, and some flexibility is permitted, provided that the manufacturer's selection of test samples is designed to give proper representation and the minimum test quantities are met.

Where a major change in types made by a factory creates a situation where the minimum test quantities are not met in the 12-month period, it shall be sufficient to show that the rate of testing at the time was compatible with the requirements of this clause.

12.3 Sampling for Batch Testing

12.3.1 There shall be selected a random sample for an ITQ consisting of 50 lamps.

12.3.2 There shall be selected at random an RTQ comprising 100 lamps. The ITQ can be used as part of the RTQ.

12.3.3 From lamps, which have passed the rating test. there shall be selected at random an LTQ of 50 lamps.

Mandatory or voluntary standards

Are there some mandatory or voluntary standards applicable ? If yes, which ones ?

Please indicate the reference of the applicable standards

IS 10322-5-5 (1987) : Luminaires, Part 5: Particular requirements, Section 5: Flood Light (Superseding IS:1947) [ETD 24: Illumination Engineering and Luminaries]

i.e Indian Standard – SPECIFICATION FOR LUMINAIRES -Part 5 Particular Requirements-Section 5 Flood-Lights

4. CLASSIFICATION OF LUMINAIRES

4.1 Flood-lights shall be classified in accordance with the provisions of 5 of Part 1 of this standard.

6. CONSTRUCTION

6.0 The provisions of Part 2 of this standard shall apply together with the requirements of 6.1 to 6.6.

6.1 Floodlights for use outdoors shall have protection against the ingress of moisture at least equivalent to IP x 3.

6.2 Lampholder brackets and lamp supports, where used, shall withstand normal usage throughout the life of the floodlight. They shall accept and retain lamps which are within the dimensional tolerances stated in the appropriate Indian standards where applicable, and locate the lamp or lamps in the designed relationship to the optical control devices in the floodlight.

6.3 When provision is made for alternative sizes of lamps or light centre positions, the adjusting means shall be positively and firmly retained in the selected position.

6.4 Refractors, reflectors or any other light controlling components shall be so marked or constructed that they can be fitted or replaced only in the correct relationship to the light source.

6.5 The means for attaching the floodlight to its support shall be appropriate to the weight of the floodlight.

For flood-lights for use above ground level outdoors, the mounting arrangement shall be such as to make the flood-light connection withstand wind speeds of 150 km/h on the projected surface of the floodlight assembly without undue deflection.

Fixings which carry the weight of the flood-light and internal accessories shall be provided with appropriate means to prevent the dislodging of any part of the floodlight by vibration, either in service or during maintenance.

Parts of flood-lights for mounting heights 3 m or higher which are fixed other than with at least two devices, for example, screws or equivalent means of sufficient strength, shall have such extra protection as to prevent those parts falling and endangering persons, animals and surroundings, if a fixing device fails under normal conditions.

Compliance shall be checked by inspection and, for flood-lights for use above ground level outdoors, by the additional test 6.5.1.

6.5.1 The flood-light is mounted with its largest projected area as viewed in elevation lying in the horizontal plane, and with the means of attachment secured in accordance with the manufacturer's recommendations.

For Flood-lights for use above ground level outdoors, a constant evenly distributed load is applied for 10 min on the floodlight using sand bags providing 2.4 kN m sq. of floodlight projected area. The flood-light is then turned 180 degree, in the vertical plane, about the point of attachment and the test is repeated.

During the test, there shall be no failure or movement about the point of attachment, and after either part of this test there shall not be permanent set exceeding 1 degree.

6.6 Where means for angular adjustment are provided, there shall be provision for positive locking after any such adjustments have been effected.

6.7 Flood-lights for use outdoors shall be resistant to vibrations which may occur during normal use.

7. CREEPAGE DISTANCES AND CLEARANCES

7.1 The provisions given in 4 of Part 4 of this standard shall apply.

8. PROVISIONS FOR EARTHING

8.1 The provisions given in 2.0 of Part 2 of this standard shall apply.

9. TERMINALS

9.1 The provisions given in Part 2 of this standard shall apply.

10. EXTERNAL AND INTERNAL WIRING

10.1 The provisions given in 19 of Part 2 of this standard shall apply.

11. PROTECTION AGAINST ELECTRIC SHOCK

11.1 The provisions given in 21 of Part 2 of this standard shall apply.

12. PHOTOMETRIC REQUIREMENTS

12.1 The manufacturer or supplier shall provide the following data relating to performance of the luminaires:

- a) Luminous intensity curves for both vertical and horizontal planes,
- b) Beam spread on the basis of $I > 0.5 I_{\max}$ and $I > 0.1 I_{\max}$ for both vertical and horizontal planes,
- c) Maximum intensity, and
- d) Total light output ratio.

12.1.1 The manufacturer or supplier shall provide the following data relating to performance of the luminaire on request:

- a) Relative insolux diagram for at least 0 degree aiming angle,
- b) Relative isocandela diagram, and
- c) Data on zonal flux.

12.2 Light Controlling Components – The attachment of reflectors or any other light controlling components shall be such that they can only be fitted or replaced in the correct relationship to their light source.

IS 418:2004 i.e Indian Standard-TUNGSTEN FILAMENT LAMPS FOR DOMESTIC AND SIMILAR GENERAL LIGHTING PURPOSES (Fourth Revision)

4 LAMP CHARACTERISTICS AND SPECIFICATIONS

4.1 Lamp characteristics and specific performance limits are listed on the individual lamp data sheets. These data sheets are given in Annex G.

4.2 Each lamp data sheet defines a particular lamp "group" by listing the characteristics and limiting values that apply. The technical specifications on each sheet are: dimensions, minimum rated luminous flux, lumen maintenance, rated life, and information for luminaries design.

4.3 The sequence of the data sheets in Annex G is by wattage within the following sub-divisions:

Category	Datasheet Numbers
Lamps with B22 caps, rated life 1000 h	4000-4999
Lamps with E27 caps, rated life 1000 h	5000-5999

4.4 Numbering System for Lamp Data Sheets

A lamp data sheet number is made up of four parts as follows:

- the first number represents the number of this standard (IS 4 18);
- The second part is the letter grouping 'IS' (Indian Standard);

The third part is the basic data sheet number from the series in 4.3;

- the fourth part is a number indicating the edition of the sheet.

NOTE — When amendments are made to data sheets, the affected pages will be issued with an updated edition number for example, if data sheet 418-IS-4005-1 were amended, the new issue would be numbered 418-IS-4005-2.

5 GENERAL REQUIREMENTS

5.1 The lamps on which compliance with this standard is claimed shall comply with the requirements of Doc ET 23 (5291).

5.2 Lamps shall be so designed that their performance is reliable in normal and accepted use. In general, this can be achieved by satisfying the requirements of 6 to 10.

5.3 Lamps shall be tested under the procedures of Annex A on test procedure.

NOTE — The life test may be carried out as per the alternative method given in Annex J This alternative method shall be valid up to 31 December 2000

5.4 For the guidance to the lamp manufacturers, the requirements for glass shells are given in IS 1112 (Part 1) and IS 12897 for lamp filament.

7 LAMP DIMENSIONS

7.1 Lamps shall comply with the dimensional requirements specified on the appropriate lamp data sheet.

7.2 Lamp with E27 caps shall comply with the gauge for testing contact making as given in data sheet 7006-50.

8 CHARACTERISTICS AND TOLERANCES OF INITIAL READINGS

8.1 Wattage

The initial wattage of individual lamps shall not exceed 104 percent of the rated wattage specified on the relevant lamp data sheet plus 0.5 W.

8.2 Luminous Flux Initial

8.2.1 Rated luminous flux of the lamps shall not be less than the values shown on the relevant lamp data sheet.

8.2.2 The initial luminous flux readings of individual frosted, frosted equivalently coated or clear lamps shall not be less than 93 percent of the rated luminous flux.

9 LUMEN MAINTENANCE

9.1 The lumen maintenance of individual lamps at 75 percent of rated life shall be not less than the minimum value specified on the relevant lamp data sheet.

NOTE — For the compliance conditions of 11.1.2.6 and 11.2.3 lamps that do not satisfy these requirements are treated as life failures

10 LIFE TEST REQUIREMENTS

10.1 The truncated average life of a normal life test or the equivalent truncated average life of an accelerated life test, calculated by the method of B.1.1 of Annex B, shall be equal to or greater than the limits in B.1.2. as related to rated life and the LTQ.

10.2 Individual lamps shall have a life of not less than 70 percent of rated life.

11 CONDITIONS OF COMPLIANCE

11.1 Whole Production of a Manufacturer

Compliance is proven by satisfying the requirements of 5 to 10 (general, dimensional, electrical, photometric, and life requirements) assessed on the following basis.

11.1.1 Pre-Compliance Testing for Certification Purposes (Applicable for Manufacturers)

NOTE — For certification purposes a recommended pre-compliance test is given in Annex C Such a test provides temporary recognition of a supplier as explained in C-1.

11.1.2 Compliance of Manufacturer's Test Data

11.1.2.1 The assessment shall be based on the test data in the manufacturer's records grouped together, meeting the requirements of 11.1.2.3.

11.1.2.2 For certification purposes, the manufacturer shall declare a list of lamp types which are to be within the scope of this standard, and this shall be taken to include all lamps so listed made by the manufacturer. Notifications of additions or deletions may be made at any time

11.1.2.3 The whole production of a manufacturer shall be considered as satisfying the requirements of this standard if the conditions of 11.1.2.4, 11.1.2.5 and 11.1.2.6 are fulfilled for at least 75 percent (rounded to the nearest whole number) of the total number of types, as selected in 12.2.2 for which test data has been submitted.

11.1.2.4 Dimensions

A type from the whole production of a manufacturer shall be considered to comply if, for that type, the number of lamps in the manufacturer's records failing the dimensional requirements of 7 does not exceed the qualifying limit shown in Table D-1, Annex D. (This number of lamps is established from data supplied by the manufacturer).

11.1.2.5 Initial readings

A type from the whole production of a manufacturer shall be considered to comply with the initial requirements, if for that type

a) the number of lamps in the manufacturer's records whose wattage is above the limitation of 8.1 does not exceed the value given in Table D-2, Annex D;

b) the number of lamps in the manufacturer's records having luminous flux values below the limitation of 8.2.2, does not exceed the value given in Table D-2, Annex D.

11.1.2.6 Life and lumen maintenance

A type from the whole production of a manufacturer shall be considered to comply if for that type:

a) the manufacturer's records show that the truncated average life results satisfy the requirements of 10.1; and

b) the total number of individual lamps failing the requirements of 10.2 together with those failing in 9.1 does not exceed the number given in Table D-3, Annex D.

11.1.2.7 A manufacturer who has met, but no longer meets, the specified acceptance levels for 11.1.2.4, 11.1.2.5 and 11.1.2.6 shall not be disqualified from claiming compliance with this standard providing he can show that either:

a) action was taken to remedy the situation as soon as the trend was reasonably confirmed from his data and the specified acceptance level was re-established within a period of six months. When corrective action has been taken, compliance is assessed excluding the test records for the period of non-compliance from the 12-month summation. Such data shall form part of the record;

b) or the type which does not meet the specified acceptance level is deleted from the list of lamp types which he may claim are in conformity with this standard.

11.1.2.8 In the case of a lamp type, which has been deleted under 11.1.2.7 from the list (see 11.1.2.2), it may be reinstated if satisfactory results are obtained from tests on a number of lamps equivalent to the minimum 12-month period sample in the clause for which failure occurred. The sample may be collected over a short period.

11.2 Compliance of Individual Batches

Sampling for batch shall be made in accordance with 12.3. A batch shall be considered as satisfying the requirements of this standard, if the conditions contained in 11.2.1, 11.2.2 and 11.2.3 are fulfilled. If the batch fails to satisfy any of these requirements, it shall be deemed as not complying with the standard.

11.2.1 Dimensions

A batch shall be considered to comply to the dimensional requirements if the number of lamps failing 7 does not exceed four.

11.2.2 Initial Readings

A batch shall be considered to comply to initial requirements if:

- a) the number of lamps whose wattage is above the maximum value specified in 8.1 does not exceed 12;
- b) the number of lamps whose lumen are below the minimum values specified in 8.2.2 does not exceed 12.

11.2.3 Life and Lumen Maintenance

A batch shall be considered to comply to life requirements if:

- 1) the truncated average life of the LTQ satisfies 10.1;
- 2) the total number of individual lamps failing the requirement of 10.2 together with those failing the requirements given in 9 does not exceed eight.

11.2.4 Summary of Compliance Conditions for Individual Batches

A summary of the above conditions is given in the following table.

Characteristics	Sample Size	Qualifying Limit
	n	c
ITQ Dimensional requirements	50	4
RTQ Wattage	100	12
RTQ Luminous flux	100	12
LTQ Average life	50	98 percent of rated life

NOTE The sample size for LTQ shall be 20 and qualifying limit for average life and life are 960 hours and 4 hours respectively when tested as per Annex J

13 PRINCIPLES OF DIMENSIONING

13.1 Principles of Dimensioning Incandescent Lamps with Bulb Shape A, K, M or PS and Cap B22d

13.1.1 All dimensions specified on the lamp data sheets are in millimetres (mm).

13.1.2 Figure 1 presents graphical definitions of the dimensional codes for B22d capped lamps. (Please refer to PDF File attached For Figure)

13.1.3 In the bulb designations of the lamp data sheets, the numerals shown indicate the nominal bulb diameter and are not to be used for assessing the dimensions of the lamps.

13.2 Principles of Dimensioning Incandescent Lamps with Bulb Shape A, K, M or PS, and Edison Screw Cap

13.2.1 All dimensions specified on the lamp data sheets are in millimetres (mm).

13.2.2 Figure 2 presents graphical definitions of the dimensional codes for Edison screw capped lamps. (Please refer to PDF File attached For Figure)

13.2.3 In the bulb designations of the lamp data sheets, the numerals shown indicate the nominal bulb diameter and are not to be used for assessing the dimensions of the lamps.

ANNEX A (Clause 5.3) TEST PROCEDURE (Please refer to PDF File attached)

ANNEX B (Clause 10.1) LIFE CALCULATION AND LIMITS (Please refer to PDF File attached)

ANNEX C (Clauses 11.1.1 and 12.2.1) RECOMMENDED PRE-COMPLIANCE TESTS FOR CERTIFICATION PURPOSES (APPLICABLE FOR MANUFACTURERS) (Please refer to PDF File attached)

ANNEX D (Clauses 11.1.2.4, 11.1.2.5 and 11.1.2.6) STATISTICAL COMPLIANCE TABLES (Please refer to PDF File attached)

ANNEX E (Tables D-1, D-2 and D-3) STATISTICAL CONCEPTS AND BASIS OF THIS STANDARD (Please refer to PDF File attached)

ANNEX F (Clauses A-4.7 and A-4.9) TEST RACK CIRCUIT CHARACTERISTICS (Please refer to PDF File attached)

ANNEX G (Clauses 1.1, 4.1 and 4.3) GROUPING OF LAMP DATA SHEETS (Please refer to PDF File attached)

ANNEX H (Clause 3.21) REFERENCE LAMPS (Please refer to PDF File attached)

IS 2418-2 (1977): Tubular Fluorescent Lamps for General Lighting Service, Part II: Standard Lamp Data Sheets [ETD 23: Electric Lamps and their Auxiliaries]

3. GENERAL PRINCIPLE OF PRESENTATION OF DATA SHEETS

3.0 This standard will be in a leaf form to permit the easy insertion of additional and/or revised data sheets.

3.1 Numbering of Data Sheets – Each data sheet shall be numbered in the following manner;

2418/0000-x

Where

X is Edition Number.

3.1.1 Grouping

Sheet Numbers		Method of Starting		Cathode
Type			Resistance	
2418/1000-1999	Starter		Preheated	-
2418/2000-2999	Starter		Preheated	-
2418/3000-3999				
2418/4000-4999	Starterless		Preheated	High
2418/5000-5999	Starterless		Preheated	Low

3.2 List of Standard Lamp Data Sheets

Lamp Rating	Cap	Method of starting		Cathode	Sheet No.
Type			Resistance		
4w	G-5	Starter	Preheated	-	2418/1020
6W	G-5	Starter	Preheated	-	2418/1030
8W	G-5	Starter	Preheated	-	2418/1040
13W	G-13	Starter	Preheated	-	2418/1050
20W	G-13	Starter	Preheated	-	2418/1110
40W	G-13	Starter	Preheated	-	2418/1310

65W	G-13	Starter	Preheated	-	2418/1550
80W	G-13	Starter	Preheated	-	2418/1710
40W	G-13	Starterless	Preheated	High	2418/4310
80W	G-13	Starterless	Preheated	High	2418/4710
65W	G-13	Starterless	Preheated	Low	2418/5550
18W	G-13	Starter	Preheated	-	2418/1105-1
36W	G-13	Starter	Preheated	-	2418/1305-1
58W	G-13	Starter	Preheated	-	2418/1545-1
18W HL	G-13	Starter	Preheated	-	2418/1100-1
36W HL	G-13	Starter	Preheated	-	2418/1130-1

NOTE – A lamp if marked 65/80W is required to comply with the requirements of both 65W and 80W lamps separately.

4. DIAGRAMMATIC DRAWING FOR LOCATION OF LAMP DIMENSIONS

4.1 Figure 1 is intended only to indicate dimensions to be controlled and is to be used in conjunction with the relevant map standard sheets. (Please refer to PDF file attached for Figure)

TUBULAR FLUORESCENT LAMP DATA SHEET NO> 2418/1020-1

NOMINAL DIMENSIONS	CAP	RATINGS	STARTING
(1)	(2)	(3)	(4)
mm		W	
150 X 15	G-5	4	With Starter

NOTE-This lamp is subject to type test only.

Mechanical and Physical Requirements

For lamp marking, tubes, dimensions and caps, 4 and 5 of IS:2418 (Part I)-1977 (i.e. Specification for tubular fluorescent lamps for general lighting service: Part I Requirements and Tests - First revision) shall apply.

LAMP DIMENSIONS, mm (see Fig. 1)

A	B	C	D
Max		Max	Max
(1)		(4)	(5)
Max		Min	
(2)		(3)	
135.7	142.9	140.5	150.0
			16.0

Electrical characteristics

For position of operation and ageing, 6.2 of IS : 2418 (part I)- 1977* (i.e Specification for tabular fluorescent lamps for general lighting service : Part I Requirements and test-first revision) shall apply.

LAMP ELECTRICAL CHARACTERISTICS (AT 25 Degree C AMBIENT TEMP)

RATED WATTAGE		OBJECTIVE WATTAGE		STARTING TEST	VOLTAGE (rms) AT LAMP TERMINALS		NOMINAL CURRENT
				VOLTAGE			
Obj		Max		Min	Running	Preheat	
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
W	W	V	V	V	V	A	A
4.0	4.5	90	29	32	26	0.170	0.200

Ballast Characteristics

The requirements of IS :1534 (Part I)-1967 (i.e Specification for ballasts for fluorescent lamps: Part I for Switch start circuits -First revision)

REFERENCE BALLAST CHARACTERISTICS

RATED WATTAGE	RATED VOLTAGE	CALIBRATION CUR- RENT	RATIO VOLTAGE CUR- RENT	POWER FACTOR
(1)	(2)	(3)	(4)	(5)
W	V	A	Ohm	
6	127	0.160	700	0.12

TUBULAR FLUORESCENT LAMP DATA SHEET NO. 2418/1030-1

NOMINAL DIMENSIONS	CAP	RATING	STARTING
(1)	(2)	(3)	(4)
mm		W	
225 X 15	G-15	6	With Starter

NOTE – This lamp is subject to type test only.

Mechanical and Physical requirements

For lamp marking, tubes, dimensions and caps, 4 and 5 of IS:2418 (Part I)-1977(i.e Specification for tabular fluorescent lamps for general lighting service : Part I Requirements and test-first revision) shall apply.

LAMP DIMENSIONS, mm (see Fig. 1)

A	B	C	D
Max		Max	Max
(1)		(4)	(5)
Max		Min	
(2)		(3)	
211.9	219.1	216.7	226.2
			16.0

Electrical characteristics

For position of operation and ageing, 6.2 of IS : 2418 (Part I)- 1977* (i.e Specification for tabular fluorescent lamps for general lighting service : Part I Requirements and test-first revision) shall apply.

LAMP ELECTRICAL CHARACTERISTICS (AT 25 Degree C AMBIENT TEMP)

RATED WATTAGE		OBJECTIVE WATTAGE		STARTING TEST	VOLTAGE (rms) AT LAMP TERMINALS		NOMINAL CURRENT
				VOLTAGE			
(1)		(2)		(3)			
Obj		Max		Min	Running	Preheat	
(4)		(5)		(6)			
(7)				(8)			
W	W	V	V	V	V	A	A
6	6.0	95	42	47	37	0.160	0.200

Ballast Characteristics

The requirements of IS :1534 (Part I)-1967 (i.e Specification for ballasts for fluorescent lamps: Part I for Switch start circuits -First revision) shall apply.

REFERENCE BALLAST CHARACTERISTICS

RATED WATTAGE	RATED VOLTAGE	CALIBRATION CUR- RENT	RATIO VOLTAGE CUR- RENT	POWER FACTOR
(1)	(2)	(3)	(4)	(5)
W	V	A	Ohm	
6	127	0.160	700	0.12

TUBULAR FLUORESCENT LAMP DATA SHEET NO. 2418/1040-1

NOMINAL DIMENSIONS	CAP	RATING	STARTING
(1)	(2)	(3)	(4)
mm		W	
300 X 15	G-15	8	With Starter

NOTE – This lamp is subject to type test only.

Mechanical and Physical requirements

For lamp marking, tubes, dimensions and caps, 4 and 5 of IS:2418 (Part I)-1977 (i.e Specification for tabular fluorescent lamps for general lighting service : Part I Requirements and test-first revision) shall apply.

LAMP DIMENSIONS, mm (see Fig. 1)

A	B	C	D
Max		Max	Max
(1)		(4)	(5)
Max		Min	
(2)		(3)	
288.1	295.3	292.9	302.4
			16.0

Electrical characteristics

For position of operation and ageing, 6.2 of IS : 2418 (part I)- 1977* (i.e Specification for tabular fluorescent lamps for general lighting service : Part I Requirements and test-first revision) shall apply.

LAMP ELECTRICAL CHARACTERISTICS (AT 25 Degree C AMBIENT TEMP)

RATED WATTAGE		OBJECTIVE WATTAGE		STARTING TEST	VOLTAGE (rms) AT LAMP TERMINALS	NOMINAL CURRENT	
				VOLTAGE			
(1)		(2)		(3)			
Obj		Max		Min	Running	Preheat	
(4)		(5)		(6)	(7)	(8)	
W	W	V	V	V	V	A	A
8	7.2	95	57	63	51	0.145	0.200

Ballast Characteristics

The requirements of IS :1534 (Part I)-1967 (i.e Specification for ballasts for fluorescent lamps: Part I for Switch start circuits -First revision) shall apply.

REFERENCE BALLAST CHARACTERISTICS

RATED WATTAGE	RATED VOLTAGE	CALIBRATION CURRENT	RATIO VOLTAGE CURRENT	POWER FACTOR
(1)	(2)	(3)	(4)	(5)
W	V	A	Ohm	
6	127	0.160	700	0.12

TUBULAR FLUORESCENT LAMP DATA SHEET NO. 2418/1050-1

NOMINAL DIMENSIONS	CAP	RATING	STARTING
(1)	(2)	(3)	(4)
mm		W	
525 X 15	G-15	13	With Starter

NOTE – This lamp is subject to type test only.

Mechanical and Physical requirements

For lamp marking, tubes, dimensions and caps, 4 and 5 of IS:2418 (Part I)-1977 (i.e Specification for tabular fluorescent lamps for general lighting service : Part I Requirements and test-first revision) shall apply.

LAMP DIMENSIONS, mm (see Fig. 1)

A	B	C	D
Max		Max	Max
(1)		(4)	(5)
Max		Min	
(2)		(3)	
516.7	529.3	521.5	531
			16.0

Electrical characteristics

For position of operation and ageing, 6.1 of IS : 2418 (part I)- 1977* (i.e Specification for tubular fluorescent lamps for general lighting service : Part I Requirements and test-first revision) shall apply.

LAMP ELECTRICAL CHARACTERISTICS (AT 25 Degree C AMBIENT TEMP)

RATED WATTAGE		OBJECTIVE WATTAGE		STARTING TEST	VOLTAGE (rms) AT LAMP TERMINALS	NOMINAL CURRENT	
				VOLTAGE			
(1)		(2)		(3)			
Obj		Max		Min	Running	Preheat	
(4)		(5)		(6)	(7)	(8)	
W	W	V	V	V	V	A	A
13	13	180	95	105	85	0.165	0.225

Ballast Characteristics

The requirements of IS :1534 (Part I)-1967 (i.e Specification for ballasts for fluorescent lamps: Part I for Switch start circuits -First revision) shall apply.

REFERENCE BALLAST CHARACTERISTICS

RATED WATTAGE	RATED VOLTAGE	CALIBRATION CUR- RENT	RATIO VOLTAGE CUR- RENT	POWER FACTOR
(1)	(2)	(3)	(4)	(5)
W	V	A	Ohm	
13	236	0.165	1 200	0.075 + 0.005

TUBULAR FLUORESCENT LAMP DATA SHEET NO. 2418/1110-1

NOMINAL DIMENSIONS	CAP	RATING	STARTING
(1)	(2)	(3)	(4)
mm		W	
600 X 38	G-13	20	With Starter

Mechanical, Physical and Starting Requirements

For tubes, dimensions, Caps and starting requirements; Lamp marking and for conditions of compliance, 4, 5 And 7 of IS; 2418 (Part I)-1977 (i.e Specification for tubular fluorescent lamps for general lighting service: Part I Requirements and tests -First revision) shall apply.

LAMP DIMENSIONS, mm (see Fig. 1)

A	B	C	D
Max		Max	Max
(1)		(4)	(5)
Max		Min	
(2)		(3)	
589.8	596.9	594.4	604.0
			40.5

LAMP STARTING TEST

TEST VOLTAGE	BALLAST VOLATGE RATING
(1)	(2)
V	V
95	110

Initial Readings and Colour

For lamp voltage, wattage and luminous characteristics; and for conditions of compliance, 6 and 7 of IS:2418 (Part I)-1977 (i.e Specification for tubular fluorescent lamps for general lighting service: Part I Requirements and tests -first revision) shall apply.

LAMP ELECTRICAL CHARACTERISTICS (AT 25 Degree C AMBIENT TEMP)

RATED WATTAGE	OBJECTIVE WATTAGE		VOLTAGE (rms) AT LAMP TERMINALS	NOMINAL CURRENT
(1)	(2)			
Obj	Max	Min	Running	Preheat
(3)	(4)	(5)	(6)	(7)

W	W	V	V	V	A	A
20	19.3	57	64	50	0.37	0.55

LUMINOUS AND COLOUR CHARACTERISTICS

RATED COLOUR

(1)

RATED LUMINOUS FLUX

(2)

Lumens

6 500 K

940

4 300 K

1 020

2 900 K

1 060

Life Test

Minimum Value:

At 2 000 h 85 percent

At 70 percent life 80 percent

For life test requirements and for conditions of compliance 6.9 and 7 of IS: 2418 (Part I)-1977 (i.e Specification for tubular fluorescent lamps for general lighting service: Part I Requirements and tests- first revision) shall apply.

Ballast

The requirements of IS:1534 (Part I)-1967 (i.e Specification for ballasts for fluorescent lamps: Part 1 For switch start circuits –first revision) shall apply.

REFERENACE BALLAST CHARACTERISTICS

RATED WATTAGE	RATED VOLTAGE	CALIBRATION CUR- RENT	RATIO VOLTAGE CUR- RENT	POWER FACTOR
(1)	(2)	(3)	(4)	(5)
W	V	A	Ohm	
20	127	0.37	270	0.12

INFORMATION FOR BALLAST DESIGN

OPEN CIRCUIT VOLATAGE		PREHEATING CURRENT		MAXIMUM (rms) VOLTAGE ACROSS STARTER TERMINALS WITH LAMP OPERATING		EQUIVALENT RESISTANOE OF BOTH CATHODES IN SERIES	
						(6)	
				(5)			
Min (rms) at Starter		Max (Peak) at Lamp Terminals		Min		Max	
(1)		(2)		(3)		(4)	
V	V	A	A	V		Ohms	
95	400	0.333	0.800	68		50	

TUBULAR FLUORESCENT LAMP DATA SHEET NO. 2418/1310-1

NOMINAL DIMENSIONS	CAP	RATING	STARTING
(1)	(2)	(3)	(4)
mm		W	
1 200 X 38	G-13	40	With Starter

Mechanical, Physical and Starting Requirements

For tubes, dimensions, caps and starting requirements; lamp marking; and for conditions of compliance, 4, 5 And7 of IS: 2418 (Part I)-1977 (i.e Specification for tabular fluorescent lamps for general lighting service: Part I Requirements and tests -First revision) shall apply.

LAMP DIMENSIONS, mm (see Fig. 1)

A	B	C	D
Max		Max	Max
(1)		(4)	(5)

Max			Min	
(2)			(3)	
1 199.4	1 206.5	1 204.0	1 213.6	40.5

LAMP STARTING TEST

TEST VOLTAGE	BALLAST VOLATGE RATING
(1)	(2)
V	V
180	220

Initial Readings and Colour

For lamp voltage, wattage and luminous characteristics; and for conditions of compliance, 6 and 7 of IS:2418 (Part I)-1977 (i.e Specification for tubular fluorescent lamps for general lighting service: Part I Requirements and tests -first revision) shall apply.

LAMP ELECTRICAL CHARACTERISTICS (AT 25 degree C AMBIENT TEMP)

RATED WATTAGE		OBJECTIVE WATTAGE		VOLTAGE (rms) AT LAMP TERMINALS	NOMINAL CURRENT	
(1)		(2)				
Obj	Max	Min		Running	Preheat	
(3)	(4)	(5)		(6)	(7)	
W	W	V	V	V	A	A
40	39.5	103	113	93	0.43	0.65

LUMINOUS AND COLOUR CHARACTERISTICS

RATED COLOUR	RATED LUMINOUS FLUX
(1)	(2)
	Lumens
6 500 K	2 450

4 300 K	2 700
2 900 K	2 800

Life Test

Minimum Value:

At 2 000 h 85 percent

At 70 percent life 80 percent

For life test requirements and for conditions of compliance 6.9 and 7 of IS: 2418 (Part I)-1977 (i.e Specification for tubular fluorescent lamps for general lighting service: Part I Requirements and tests- first revision) shall apply.

Ballast

The requirements of IS:1534 (Part I)-1967 (i.e Specification for ballasts for fluorescent lamps: Part 1 For switch start circuits –first revision) shall apply.

REFERENACE BALLAST CHARACTERISTICS

RATED WATTAGE	RATED VOLTAGE	CALIBRATION CUR- RENT	RATIO VOLTAGE CUR- RENT	POWER FACTOR
(1)	(2)	(3)	(4)	(5)
W	V	A	Ohm	
40	220	0.43	390	0.10

INFORMATION FOR BALLAST DESIGN

OPEN CIRCUIT VOLATAGE	PREHEATING CURRENT	MAXIMUM (rms) VOLTAGE ACROSS STARTER TERMI- NALS WITH LAMP OPERAT- ING	EQUIVALENT RESISTANOE OF BOTH CATHODES IN SE- RIES
		(5)	(6)
Min (rms) at Starter	Max (Peak) at Lamp Terminals	Min	Max
(1)	(2)	(3)	(4)

V	V	A	A	V	Ohms
180	400	0.387	0.904	128	40

TUBULAR FLUORESCENT LAMP DATA SHEET NO. 2418/1550-1

NOMINAL DIMENSIONS	CAP	RATING	STARTING
(1)	(2)	(3)	(4)
mm		W	
1 500 X 38	G-13	65	With Starter

Mechanical, Physical and Starting Requirements

For tubes, dimensions, caps and starting requirements; lamp marking; and for conditions of compliance, 4, 5 And7 of IS: 2418 (Part I)-1977 (i.e Specification for tabular fluorescent lamps for general lighting service: Part I Requirements and tests -First revision) shall apply.

LAMP DIMENSIONS, mm (see Fig. 1)

A	B	C	D
Max		Max	Max
(1)		(4)	(5)
Max		Min	
(2)		(3)	
1 500.0	1 507.1	1 504.7	1 514.2
			40.5

LAMP STARTING TEST

TEST VOLTAGE	BALLAST VOLATGE RATING
(1)	(2)
V	V
180	220

Initial Readings and Colour

For lamp voltage, wattage and luminous characteristics; and for conditions of compliance, 6 and 7 of IS:2418 (Part I)-1977 (i.e Specification for tubular fluorescent lamps for general lighting service: Part I Requirements and tests -first revision) shall apply.

LAMP ELECTRICAL CHARACTERISTICS (AT 25 degree C AMBIENT TEMP)

RATED WATTAGE		OBJECTIVE WATTAGE		VOLTAGE (rms) AT LAMP TERMINALS	NOMINAL CURRENT	
(1)		(2)				
Obj		Max	Min	Running	Preheat	
(3)		(4)	(5)	(6)	(7)	
W	W	V	V	V	A	A
65	64	110	120	100	0.67	1.0

LUMINOUS AND COLOUR CHARACTERISTICS

RATED COLOUR	RATED LUMINOUS FLUX
(1)	(2)
	Lumens
6 500 K	3 930
4 300 K	4 400
2 900 K	4 600

Life Test

Minimum Value:

At 2 000 h 85 percent

At 70 percent life 80 percent

For life test requirements and for conditions of compliance 6.9 and 7 of IS: 2418 (Part I)-1977 (i.e Specification for tubular fluorescent lamps for general lighting service: Part I Requirements and tests- first revision) shall apply.

Ballast

The requirements of IS:1534 (Part I)-1967 (i.e Specification for ballasts for fluorescent lamps: Part 1 For switch start circuits –first revision) shall apply.

REFERENACE BALLAST CHARACTERISTICS

RATED WATTAGE	RATED VOLTAGE	CALIBRATION CUR- RENT	RATIO VOLTAGE CUR- RENT	POWER FACTOR
(1)	(2)	(3)	(4)	(5)
W	V	A	Ohm	
65	220	0.67	240	0.10

INFORMATION FOR BALLAST DESIGN

OPEN CIRCUIT VOLATAGE		PREHEATING CURRENT	MAXIMUM (rms) VOLTAGE ACROSS STARTER TERMI- NALS WITH LAMP OPERAT- ING		EQUIVALENT RESISTANOE OF BOTH CATHODES IN SE- RIES
			(5)	(6)	
Min (rms) at Starter		Max (Peak) at Lamp Terminals Min		Max	
(1)		(2)		(3)	
				(4)	
V	V	A	A	V	Ohms
180	400	0.603	1.41	132	25

TUBULAR FLUORESCENT LAMP DATA SHEET NO. 2418/1550-1

NOMINAL DIMENSIONS	CAP	RATING	STARTING
(1)	(2)	(3)	(4)
mm		W	
1 500 X 38	G-13	80	With Starter

Mechanical, Physical and Starting Requirements

For tubes, dimensions, caps and starting requirements; lamp marking and for conditions of compliance, 4, 5 And7 of IS: 2418 (Part I)-1977 (i.e Specification for tabular fluorescent lamps for general lighting service: Part I Requirements and tests -First revision) shall apply.

LAMP DIMENSIONS, mm (see Fig. 1)

A	B		C	D	
Max			Max	Max	
(1)			(4)	(5)	
Max			Min		
(2)			(3)		
1 500.0	1 507.1	1 504.7	1 514.2	40.5	

LAMP STARTING TEST

TEST VOLTAGE	BALLAST VOLATGE RATING
(1)	(2)
V	V
180	240

Initial Readings and Colour

For lamp voltage, wattage and luminous characteristics; and for conditions of compliance, 6 and 7 of IS:2418 (Part I)-1977 (i.e Specification for tubular fluorescent lamps for general lighting service: Part I Requirements and tests -first revision) shall apply.

LAMP ELECTRICAL CHARACTERISTICS (AT 25 degree C AMBIENT TEMP)

RATED WATTAGE		OBJECTIVE WATTAGE		VOLTAGE (rms) AT LAMP TERMINALS	NOMINAL CURRENT	
(1)		(2)				
Obj	Max	Min		Running	Preheat	
(3)	(4)	(5)		(6)	(7)	
W	W	V	V	V	A	A
80	76	99	109	89	0.87	1.30

LUMINOUS AND COLOUR CHARACTERISTICS

RATED COLOUR

(1)

6 500 K

4 300 K

2 900 K

RATED LUMINOUS FLUX

(2)

Lumens

4 330

5 000

5 200

Life Test

Minimum Value:

At 2 000 h 85 percent

At 70 percent life 80 percent

For life test requirements and for conditions of compliance 6.9 and 7 of IS: 2418 (Part I)-1977 (i.e Specification for tubular fluorescent lamps for general lighting service: Part I Requirements and tests- first revision) shall apply.

Ballast

The requirements of IS:1534 (Part I)-1967 (i.e Specification for ballasts for fluorescent lamps: Part 1 For switch start circuits –first revision) shall apply.

REFERENACE BALLAST CHARACTERISTICS

RATED WATTAGE

(1)

W

80

RATED VOLTAGE

(2)

V

240

CALIBRATION CUR- RENT

(3)

A

0.865

RATIO VOLTAGE CUR- RENT POWER FACTOR

(4)

Ohm

223

(5)

0.06

INFORMATION FOR BALLAST DESIGN

OPEN CIRCUIT VOLATAGE		PREHEATING CURRENT		MAXIMUM (rms) VOLTAGE ACROSS STARTER TERMINALS WITH LAMP OPERATING		EQUIVALENT RESISTANOE OF BOTH CATHODES IN SERIES	
						(6)	
				(5)			
Min (rms) at Starter		Max (Peak) at Lamp Terminals		Min		Max	
(1)		(2)		(3)		(4)	
V	V	A	A	V	Ohms		
180	400	0.79	1.83	128	25		

TUBULAR FLUORESCENT LAMP DATA SHEET NO. 2418/4310-1

NOMINAL DIMENSIONS	CAP	RATING	STARTING	CATHODE
(1)	(2)	(3)	(4)	(5)
mm		W		
1 200 X 38	G-13	40	Starterless	High Resistance

Mechanical, Physical and Starting Requirements

For tubes, dimensions, caps and starting requirements; lamp marking and for conditions of compliance, 4, 5 And7 of IS: 2418 (Part I)-1977 (i.e Specification for tabular fluorescent lamps for general lighting service: Part I Requirements and tests -First revision) shall apply.

LAMP DIMENSIONS, mm (see Fig. 1)

A	B	C	D
Max		Max	Max
(1)		(4)	(5)
Max		Min	
(2)		(3)	
1 199.4	1 206.5	1 204.0	1 213.6
			40.5

LAMP STARTING TEST

O.C. VOLTAGE AT LAMP TERMINALS FOR STARTING TESTBALLAST RATING FOR STARTING TEST

(1)		(3)	
Wattage		Voltage	
(2)		(3)	
V	W	V	
205	40	220	

Initial Readings and Colour

For lamp voltage, wattage and luminous characteristics; and for conditions of compliance, 6 and 7 of IS:2418 (Part I)-1977 (i.e Specification for tubular fluorescent lamps for general lighting service: Part I Requirements and tests -first revision) shall apply.

LAMP ELECTRICAL CHARACTERISTICS (AT 25 degree C AMBIENT TEMP)

RATED WATTAGE		OBJECTIVE WATTAGE		VOLTAGE (rms) AT LAMP TERMINALS		NOMINAL CURRENT
(1)		(2)		(6)		
Obj		Max		Min		
(3)		(4)		(5)		
W	W	V	V	V	A	
40	39.5	103	113	93	0.43	

LUMINOUS AND COLOUR CHARACTERISTICS

RATED COLOUR		RATED LUMINOUS FLUX	
(1)		(2)	
		Lumens	
6 500 K		2 400	
4 300 K		2 700	

2 900 K

2 800

CATHODE CHARACTERISTICS

CATHODE	TEST VOLTAGE	RESISTANCE OF EACH CATHODE
(1)	(2)	
	V	
Obj	Min	
(3)	(4)	
Ohm	Ohm	
High Resistance	8	20 17

Life Test

Minimum Value:

At 2 000 h 85 percent

At 70 percent life 80 percent

For life test requirements and for conditions of compliance 6.9 and 7 of IS: 2418 (Part I)-1977 (i.e Specification for tubular fluorescent lamps for general lighting service: Part I Requirements and tests- first revision) shall apply..

Ballast

The requirements of IS:1534 (Part I)-1967 (i.e Specification for ballasts for fluorescent lamps: Part 1 For switch start circuits –first revision) shall apply.

REFERENACE BALLAST CHARACTERISTICS

RATED WATTAGE	RATED VOLTAGE	CALIBRATION CUR- RENT	RATIO VOLTAGE CUR- RENT	POWER FACTOR
(1)	(2)	(3)	(4)	(5)
W	V	A	Ohm	
40	220	0.43	390	0.10

INFORMATION FOR BALLAST DESIGN

O.C. VOLATAGE TERMINALS

MAXIMUM CURRENT IN ANY LEAD TO PREHEAT CATHODE VOLTAGE CATHODE

(3)

Min (rms)	Max (Peak)	Min	Max
(1)	(2)	(4)	(5)
V	V	A	V
205	420	0.75	8

-

TUBULAR FLUORESCENT LAMP DATA SHEET NO. 2418/4710-1

NOMINAL DIMEN- SIONS	CAP	RATING	STARTING	Cathode
(1)	(2)	(3)	(4)	(5)
mm		W		
1 500 X 38	G-13	80	Starterless	High Resistance

Mechanical, Physical and Starting Requirements

For tubes, dimensions, caps and starting requirements; lamp marking; and for conditions of compliance, 4, 5 And 7 of IS: 2418 (Part I)-1977 (i.e Specification for tabular fluorescent lamps for general lighting service: Part I Requirements and tests -First revision) shall apply.

LAMP DIMENSIONS, mm (see Fig. 1)

A	B	C	D
Max		Max	Max
(1)		(4)	(5)
Max		Min	
(2)		(3)	
1 500.0	1 507.1	1 504.7	1 514.2
			40.5

LAMP STARTING TEST

O.C. VOLTAGE AT LAMP TERMINALS FOR STARTING TEST		BALLAST VOLATGE RATING	
(1)		(2)	
Wattage		Voltage	
(2)		(3)	
V	W		V
220	80		240

Initial Readings and Colour

For lamp voltage, wattage and luminous characteristics; and for conditions of compliance, 6 and 7 of IS:2418 (Part I)-1977 (i.e Specification for tubular fluorescent lamps for general lighting service: Part I Requirements and tests -first revision) shall apply.

LAMP ELECTRICAL CHARACTERISTICS (AT 25 degree C AMBIENT TEMP)

RATED WATTAGE		OBJECTIVE WATTAGE		VOLTAGE (rms) AT LAMP TERMINALS		NOMINAL CURRENT
(1)		(2)				(6)
Obj		Max			Min	
(3)		(4)			(5)	
W	W	V		V	V	A
80	76.0	99		109	89	0.87

LUMINOUS AND COLOUR CHARACTERISTICS

RATED COLOUR		MINIMUM RATED LUMINOUS FLUX	
(1)		(2)	
		Lumens	

6 500 K	4 330
4 300 K	5 000
2 900 K	5 200

CATHODE CHARACTERISTICS

CATHODE	TEST VOLTAGE	RESISTANCE OF EACH CATHODE
(1)	(2)	
	V	
Obj	Min	
(3)	(4)	
Ohm	Ohm	
High Resistance	8	12
		9

Life Test

Minimum Value:

At 2 000 h 80 percent

At 70 percent life 75 percent

For life test requirements and for conditions of compliance 6.9 and 7 of IS: 2418 (Part I)-1977 (i.e Specification for tubular fluorescent lamps for general lighting service: Part I Requirements and tests- first revision) shall apply.

Ballast

The requirements of IS:1534 (Part I)-1967 (i.e Specification for ballasts for fluorescent lamps: Part 1 For switch start circuits –first revision) shall apply.

REFERENACE BALLAST CHARACTERISTICS

RATED WATTAGE	RATED VOLTAGE	CALIBRATION CUR- RENT	RATIO VOLTAGE CUR- RENT	POWER FACTOR
(1)	(2)	(3)	(4)	(5)

W	V	A	Ohm	
80	240	0.865	223	0.06

INFORMATION FOR BALLAT DESIGN

O.C. VOLATAGE TERMINALS MAXIMUM CURRENT IN ANY LEAD TO PREHEAT CATHODE VOLTAGE CATHODE

(3)

Min (rms)	Max (Peak)	Min	Max
(1)	(2)	(4)	(5)

V	V	A	V	-
220	475	1.60	8	

TUBULAR FLUORESCENT LAMP DATA SHEET NO. 2418/5550-1

NOMINAL DIMEN- SIONS	CAP	RATING	STARTING	CATHODE
(1)	(2)	(3)	(4)	(5_
mm		W		
1 500 X 38	G-13	65	Starterless	Low Resistance

Mechanical, Physical and Starting Requirements

For tubes, dimensions, caps and starting requirements; lamp marking; and for conditions of compliance, 4, 5 And 7 of IS: 2418 (Part I)-1977 (i.e Specification for tabular fluorescent lamps for general lighting service: Part I Requirements and tests -First revision) shall apply.

LAMP DIMENSIONS, mm (see Fig. 1)

A	B	C	D
Max		Max	Max
(1)		(4)	(5)

Max				Min
(2)				(3)
1 500.0	1 507.1	1 504.7	1 514.2	40.5

LAMP STARTING TEST

O.C. VOLTAGE AT LAMP TERMINALS FOR STARTING TESTBALLAST RATING FOR STARTING TEST

(1)			
Wattage	Voltage		
(2)	(3)		
V	W	V	
220	65	220	

Initial Readings and Colour

For lamp voltage, wattage and luminous characteristics; and for conditions of compliance, 6 and 7 of IS:2418 (Part I)-1977 (i.e Specification for tubular fluorescent lamps for general lighting service: Part I Requirements and tests -first revision) shall apply.

LAMP ELECTRICAL CHARACTERISTICS (AT 25 degree C AMBIENT TEMP)

RATED WATTAGE		OBJECTIVE WATTAGE		VOLTAGE (rms) AT LAMP TERMINALS		NOMINAL CURRENT
(1)		(2)				(6)
Obj		Max		Min		
(3)		(4)		(5)		
W	W	V	V	V	A	
65	64.0	110	120	100	0.67	

LUMINOUS AND COLOUR CHARACTERISTICS

RATED COLOUR	RATED LUMINOUS FLUX
(1)	(2)
	Lumens
6 500 K	3 930
4 300 K	4 400
2 900 K	4 600

CATHODE CHARACTERISTICS

CATHODE	TEST VOLTAGE	RESISTANCE OF EACH CATHODE
(1)	(2)	
	V	
Obj	Min	
(3)	(4)	
Ohm	Ohm	
Low Resistance	3.6	6 4

Life Test

Minimum Value:

At 2 000 h 80 percent

At 70 percent life 75 percent

For life test requirements and for conditions of compliance 6.9 and 7 of IS: 2418 (Part I)-1977 (i.e Specification for tubular fluorescent lamps for general lighting service: Part I Requirements and tests- first revision) shall apply.

Ballast

The requirements of IS:1534 (Part I)-1967 (i.e Specification for ballasts for fluorescent lamps: Part 1 For switch start circuits –first revision) shall apply.

REFERENACE BALLAST CHARACTERISTICS

RATED WATTAGE	RATED VOLTAGE	CALIBRATION CUR- RENT	RATIO VOLTAGE CUR- RENT	POWER FACTOR
(1)	(2)	(3)	(4)	(5)
W	V	A	Ohm	
65	220	0.67	240	0.10

INFORMATION FOR BALLAT DESIGN

O.C. VOLATAGE TERMINALS		MAXIMUM CURRENT IN ANY LEAD TO PREHEAT CATHODE VOLTAGE CATHODE		
		(3)		
Min (rms)	Max (Peak)	Min	Max	
(1)	(2)	(4)	(5)	
V	V	A	V	-
220	475	1.10	3.05	

TUBULAR FLUORESCENT LAMP DATA SHEET NO. 2418/1105-1

NOMINAL DIMENSIONS	CAP	RATING	STARTING
(1)	(2)	(3)	(4)
mm		W	
600 X 26	G-13	18	With Starter

Mechanical, Physical and Starting Requirements

For tubes, dimensions, caps and starting requirements; lamp marking; and for conditions of compliance, 4, 5 And 7 of IS: 2418 (Part I)-1977 (i.e Specification for tabular fluorescent lamps for general lighting service: Part I Requirements and tests -First revision) shall apply.

LAMP DIMENSIONS, mm (see Fig. 1)

A	B	C	D
Max		Max	Max
(1)		(4)	(5)
Max		Min	
(2)		(3)	
589.8	596.9	594.4	604.0
			28.0

LAMP STARTING TEST

Test Voltage	BALLAST VOLTAGE RATING
(1)	(2)
V	V
95	110

Initial Readings and Colour

For lamp voltage, wattage and luminous characteristics; and for conditions of compliance, 6 and 7 of IS:2418 (Part I)-1977 (i.e Specification for tubular fluorescent lamps for general lighting service: Part I Requirements and tests -first revision) shall apply.

LAMP ELECTRICAL CHARACTERISTICS (AT 25 degree C AMBIENT TEMP)

RATED WATTAGE		OBJECTIVE WATTAGE		VOLTAGE (rms) AT LAMP TERMINALS	NOMINAL CURRENT	
(1)		(2)				
Obj		Max	Min	Running	Preheat	
(3)		(4)	(5)	(6)	(7)	
W	W	V	V	V	A	A
18	18	57	64	50	0.37	0.55

LUMINOUS AND COLOUR CHARACTERISTICS

RATED COLOUR

(1)

6 500 K

4 300 K

Life Test

Minimum value: for lumens maintenance:

At 2 000 h 80 percent

At 70 percent life 80 percent

For life test requirements and for conditions of compliance 6.9 and 7 of IS: 2418 (Part I)-1977 (i.e Specification for tubular fluorescent lamps for general lighting service: Part I Requirements and tests- first revision) shall apply.

Ballast

The requirements of IS:1534 (Part I)-1967 (i.e Specification for ballasts for fluorescent lamps: Part 1 For switch start circuits –first revision) shall apply.

REFERENACE BALLAST CHARACTERISTICS

RATED WATTAGE	RATED VOLTAGE	CALIBRATION CUR- RENT	RATIO VOLTAGE CUR- RENT	POWER FACTOR
(1)	(2)	(3)	(4)	(5)
W	V	A	Ohm	
20	127	0.37	270	0.12

INFORMATION FOR BALLAT DESIGN

OPEN CIRCUIT VOLATAGE	PREHEATING CURRENT	MAXIMUM (rms) VOLTAGE ACROSS STARTER TERMI- NALS WITH LAMPS OPERAT- ING	EQUIVALENT RESISTANCE OF BOTH CATHODE IN SE- RIES
		(5)	(6)

Min (rms) at Starter		Max (Peak) at Lamps Terminals		Min	Max
(1)	(2)			(3)	(4)
V	V	A	A		ohms
95	400	0.333	0.800	1.10	3.05

TUBULAR FLUORESCENT LAMP DATA SHEET NO. 2418/1305-1

NOMINAL DIMENSIONS	CAP	RATING	STARTING
(1)	(2)	(3)	(4)
mm		W	
1 200 X 26	G-13	36	With Starter

Mechanical, Physical and Starting Requirements

For tubes, dimensions, caps and starting requirements; lamp marking; and for conditions of compliance, 4, 5 And 7 of IS: 2418 (Part I)-1977 (i.e Specification for tubular fluorescent lamps for general lighting service: Part I Requirements and tests -First revision) shall apply.

LAMP DIMENSIONS, mm (see Fig. 1)

A	B	C	D
Max		Max	Max
(1)		(4)	(5)
Max		Min	
(2)		(3)	
1 199.4	1 206.5	1 204.0	1 213.6
			28.0

LAMP STARTING TEST

Test Voltage	BALLAST VOLTAGE RATING
(1)	(2)
V	V
185	220

Initial Readings and Colour

For lamp voltage, wattage and luminous characteristics; and for conditions of compliance, 6 and 7 of IS:2418 (Part I)-1977 (i.e Specification for tubular fluorescent lamps for general lighting service: Part I Requirements and tests -first revision) shall apply.

LAMP ELECTRICAL CHARACTERISTICS (AT 25 degree C AMBIENT TEMP)

RATED WATTAGE		OBJECTIVE WATTAGE		VOLTAGE (rms) AT LAMP TERMINALS	NOMINAL CURRENT	
(1)		(2)				
Obj		Max	Min	Running	Preheat	
(3)		(4)	(5)	(6)	(7)	
W	W	V	V	V	A	A
36	36	103	113	93	0.43	0.65

LUMINOUS AND COLOUR CHARACTERISTICS

RATED COLOUR	RATED LUMINOUS FLUX
(1)	(2)
	Lumens
6 500 K	2 245
4 300 K	2 800

Life Test

Minimum value: for lumens maintenance:

At 2 000 h 85 percent

At 70 percent life 80 percent

For life test requirements and for conditions of compliance 6.9 and 7 of IS: 2418 (Part I)-1977 (i.e Specification for tubular fluorescent lamps for general lighting service: Part I Requirements and tests- first revision) shall apply.

Ballast

The requirements of IS:1534 (Part I)-1967 (i.e Specification for ballasts for fluorescent lamps: Part 1 For switch start circuits –first revision) shall apply.

REFERENACE BALLAST CHARACTERISTICS

RATED WATTAGE	RATED VOLTAGE	CALIBRATION CUR- RENT	RATIO VOLTAGE CUR- RENT	POWER FACTOR
(1)	(2)	(3)	(4)	(5)
W	V	A	Ohm	
40	220	0.43	390	0.10

INFORMATION FOR BALLAT DESIGN

OPEN CIRCUIT VOLATAGE	PREHEATING CURRENT	MAXIMUM (rms) VOLTAGE ACROSS STARTER TERMI- NALS WITH LAMPS OPERAT- ING	EQUIVALENT RESISTANCE OF BOTH CATHODE IN SE- RIES
		(5)	(6)
Min (rms) at Starter	Max (Peak) at Lamps Terminals	Min	Max
(1)	(2)	(3)	(4)
V	V	A	A
			ohms
180	400	0.387	0.904
			128
			40

NOMINAL DIMENSIONS	CAP	RATING	STARTING
(1)	(2)	(3)	(4)
mm		W	
1 500 X 26	G-13	58	With Starter

Mechanical, Physical and Starting Requirements

For tubes, dimensions, caps and starting requirements; lamp marking; and for conditions of compliance, 4, 5 And7 of IS: 2418 (Part I)-1977 (i.e Specification for tubular fluorescent lamps for general lighting service: Part I Requirements and tests -First revision) shall apply.

LAMP DIMENSIONS, mm (see Fig. 1)

A	B	C	D
Max		Max	Max
(1)		(4)	(5)
Max		Min	
(2)		(3)	
1 500.0	1 507.1	1 504.7	1 514.2
			28

LAMP STARTING TEST

Test Voltage	BALLAST VOLTAGE RATING
(1)	(2)
V	V
180	220

Initial Readings and Colour

For lamp voltage, wattage and luminous characteristics; and for conditions of compliance, 6 and 7 of IS:2418 (Part I)-1977 (i.e Specification for tubular fluorescent lamps for general lighting service: Part I Requirements and tests -first revision) shall apply.

LAMP ELECTRICAL CHARACTERISTICS (AT 25 degree C AMBIENT TEMP)

RATED WATTAGE		OBJECTIVE WATTAGE		VOLTAGE (rms) AT LAMP TERMINALS	NOMINAL CURRENT	
(1)		(2)				
Obj		Max	Min	Running	Preheat	
(3)		(4)	(5)	(6)	(7)	
W	W	V	V	V	A	A
58	58	110	120	100	0.67	1.0

LUMINOUS AND COLOUR CHARACTERISTICS

RATED COLOUR	RATED LUMINOUS FLUX
(1)	(2)
	Lumens
6 500 K	3 930
4 300 K	4 400

Life Test

Minimum value: for lumens maintenance:

At 2 000 h 85 percent

At 70 percent life 80 percent

For life test requirements and for conditions of compliance 6.9 and 7 of IS: 2418 (Part I)-1977 (i.e Specification for tubular fluorescent lamps for general lighting service: Part I Requirements and tests- first revision) shall apply.

Ballast

The requirements of IS:1534 (Part I)-1967 (i.e Specification for ballasts for fluorescent lamps: Part 1 For switch start circuits –first revision) shall apply.

REFERENACE BALLAST CHARACTERISTICS

RATED WATTAGE	RATED VOLTAGE	CALIBRATION CUR- RENT	RATIO VOLTAGE CUR- RENT	POWER FACTOR
(1)	(2)	(3)	(4)	(5)

W	V	A	Ohm	
65	220	0.67	240	0.10

INFORMATION FOR BALLAT DESIGN

OPEN CIRCUIT VOLATAGE	PREHEATING CURRENT	MAXIMUM (rms) VOLTAGE ACROSS STARTER TERMI- NALS WITH LAMPS OPERAT- ING	EQUIVALENT RESISTANCE OF BOTH CATHODE IN SE- RIES
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		(5)	(6)
Min (rms) at Starter	Max (Peak) at Lamps Terminals	Min	Max
(1)	(2)	(3)	(4)

V	V	A	A	V	ohms
180	400	0.387	0.904	128	40

TUBULAR FLUORESCENT LAMP DATA SHEET NO. 2418/1100-1

NOMINAL DIMENSIONS	CAP	RATING	STARTING
(1)	(2)	(3)	(4)
mm		W	
600 X 26	G-13	18	With Starter

Mechanical, Physical and Starting Requirements

For tubes, dimensions, caps and starting requirements; lamp marking; and for conditions of compliance, 4, 5 And7 of IS: 2418 (Part I)-1977 (i.e Specification for tabular fluorescent lamps for general lighting service: Part I Requirements and tests -First revision) shall apply.

LAMP DIMENSIONS, mm

A	B	C	D
Max		Max	Max
(1)		(4)	(5)
Max		Min	
(2)		(3)	
589.8	596.9	594.4	604.0
			28.0

LAMP STARTING TEST

Test Voltage	BALLAST VOLTAGE RATING
(1)	(2)
V	V
95	110

Initial Readings and Colour

For lamp voltage, wattage and luminous characteristics; and for conditions of compliance, 6 and 7 of IS:2418 (Part I)-1977 (i.e Specification for tubular fluorescent lamps for general lighting service: Part I Requirements and tests -first revision) shall apply.

LAMP ELECTRICAL CHARACTERISTICS (AT 25 degree C AMBIENT Temperature)

RATED WATTAGE		OBJECTIVE WATTAGE		VOLTAGE (rms) AT LAMP TERMINALS	NOMINAL CURRENT	
(1)		(2)				
Obj		Max	Min	Running	Preheat	
(3)		(4)	(5)	(6)	(7)	
W	W	V	V	V	A	A
18	18	57	64	50	0.37	0.55

LUMINOUS AND COLOUR CHARACTERISTICS

RATED COLOUR

(1)

6 500 K

4 000 K

2 700 K

RATED LUMINOUS FLUX

(2)

Lumens

1 250

1 300

1 300

Life Test

Minimum value: for lumens maintenance:

At 2 000 h 90 percent

At 70 percent life 85 percent

For life test requirements and for conditions of compliance 6.9 and 7 of IS: 2418 (Part I)-1977 (i.e Specification for tubular fluorescent lamps for general lighting service: Part I Requirements and tests- first revision) shall apply.

Ballast

The requirements of IS:1534 (Part I)-1967 (i.e Specification for ballasts for fluorescent lamps: Part 1 For switch start circuits –first revision) shall apply.

REFERENACE BALLAST CHARACTERISTICS

RATED WATTAGE	RATED VOLTAGE	CALIBRATION CUR- RENT	RATIO VOLTAGE CUR- RENT	POWER FACTOR
(1)	(2)	(3)	(4)	(5)
W	V	A	Ohm	
20	127	0.37	270	0.12

INFORMATION FOR BALLAT DESIGN

OPEN CIRCUIT VOLATAGE		PREHEATING CURRENT		MAXIMUM (rms) VOLTAGE ACROSS STARTER TERMINALS WITH LAMPS OPERATING		EQUIVALENT RESISTANCE OF BOTH CATHODE IN SERIES	
				(5)		(6)	
Min (rms) at Starter		Max (Peak) at Lamps Terminals		Min		Max	
(1)		(2)		(3)		(4)	
V	V	A		A			ohms
95	400	0.333		0.800	68		50

TUBULAR FLUORESCENT LAMP DATA SHEET NO. 2418/1130-1

NOMINAL DIMENSIONS		CAP	RATING	STARTING
(1)		(2)	(3)	(4)
mm			W	
1 200 X 26		G-13	36	With Starter

Mechanical, Physical and Starting Requirements

For tubes, dimensions, caps and starting requirements; lamp marking; and for conditions of compliance, 4, 5 And7 of IS: 2418 (Part I)-1977 (i.e Specification for tabular fluorescent lamps for general lighting service: Part I Requirements and tests -First revision) shall apply.

LAMP DIMENSIONS, mm

A	B	C	D
Max		Max	Max
(1)		(4)	(5)
Max		Min	
(2)		(3)	

1 199.4	1 206.5	1 204.0	1 213.6	28.0
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LAMP STARTING TEST

Test Voltage	BALLAST VOLTAGE RATING
(1)	(2)
V	V
180	220

Initial Readings and Colour

For lamp voltage, wattage and luminous characteristics; and for conditions of compliance, 6 and 7 of IS:2418 (Part I)-1977 (i.e Specification for tubular fluorescent lamps for general lighting service: Part I Requirements and tests -first revision) shall apply.

LAMP ELECTRICAL CHARACTERISTICS (AT 25 degree C Ambient Temperature)

RATED WATTAGE		OBJECTIVE WATTAGE		VOLTAGE (rms) AT LAMP TERMINALS	NOMINAL CURRENT	
(1)		(2)				
Obj	Max	Min		Running	Preheat	
(3)	(4)	(5)		(6)	(7)	
W	W	V	V	V	A	A
36	36	103	113	93	0.43	0.65

LUMINOUS AND COLOUR CHARACTERISTICS

RATED COLOUR	RATED LUMINOUS FLUX
(1)	(2)
	Lumens
6 500 K	3 150
4 000 K	3 200
2 700 K	3 200

Life Test

Minimum value: for lumens maintenance:

At 2 000 h 90 percent

At 70 percent life 85 percent

For life test requirements and for conditions of compliance 6.9 and 7 of IS: 2418 (Part I)-1977 (i.e Specification for tubular fluorescent lamps for general lighting service: Part I Requirements and tests- first revision) shall apply.

Ballast

The requirements of IS:1534 (Part I)-1967 (i.e Specification for ballasts for fluorescent lamps: Part 1 For switch start circuits –first revision) shall apply.

REFERENACE BALLAST CHARACTERISTICS

RATED WATTAGE	RATED VOLTAGE	CALIBRATION CUR- RENT	RATIO VOLTAGE CUR- RENT	POWER FACTOR
(1)	(2)	(3)	(4)	(5)
W	V	A	Ohm	
40	220	0.43	390	0.10

INFORMATION FOR BALLAT DESIGN

OPEN CIRCUIT VOLATAGE	PREHEATING CURRENT	MAXIMUM (rms) VOLTAGE ACROSS STARTER TERMI- NALS WITH LAMPS OPERAT- ING	EQUIVALENT RESISTANCE OF BOTH CATHODE IN SE- RIES
		(5)	(6)
Min (rms) at Starter	Max (Peak) at Lamps Terminals	Min	Max
(1)	(2)	(3)	(4)
V	V	A	ohms
180	400	0.387	0.904
			128
			40

Physical and mechanicals properties

What are the physical and mechanicals properties required ?

Refer to question of "Mandatory Standards" in the Technical Section.

Electrical aspects

What are requirements regarding electrical aspects ?

Refer to question of "Mandatory Standards" in the Technical Section.

Inflammability

Are there any specific requirements to ensure inflammability?

Not Applicable

Hygiene

What are the rules regarding hygiene ?

Not Applicable

Radioactivity

What are the rules regarding radioactivity ?

Not Applicable

Radiofrequency (2G,3G,4G,5G, Bluetooth, Ant+, GPS, Wifi...)

What are the rules regarding the radiofrequency?

Products making noise

What are the requirements regarding products making noise ?

Not Applicable

Chemicals substances

What are the requirements regarding chemicals substances ? Which ones are forbidden ? Limited ? Do we have to clearly indicate all substances that are in the product, or is the indication of major substances enough?"

Refer to question of "Mandatory Standards" in the Technical Section.

Packaging requirements

Can you please indicate if there are some requirements regarding the packaging of our products (mainly they are in paper, carton, plastic, textile) ? If yes, please explain them.

Not Applicable

Laboratory

In which laboratory can we perform the testing of the product according to the standard?

Please refer to following pdf files uploaded :

Group 1 List of BIS Recognised Laboratories as on 13-11-2019

Group 2 Lab of National Repute and Eminence, Facilities

Sanctions / Fines

What are the sanctions / fines in case of non conformity ?

Not Applicable

Comments

Comments

Certifications / Registrations

Certifications or registrations

Are there some certifications or registrations applicable to the product ?

Products falling under 302-1:2008 - Safety of Household and Similar Electrical Appliances are required to have the ISI Mark under Scheme I (ISI Mark) of BIS.

Products falling under IS 10322 (Part 5 / Section 6) - Luminaries (particularly relating to LED hand lamps) are required to register under Scheme II (Registration scheme) as per BIS.

Laboratory

In case of mandatory certification/ registration that we must follow please precise the name and address of any laboratories where we can conduct them

Comments

Comments

The steps of getting ISI Certificate are as below -:

Step 1 : Confirmation of Product Quality as per concerned IS Standard

- Identification of ISI Standard Code
- Collection of the Testing Sample from applicant
- Testing of sample at our lab
- Issuing the guideline about Non-Conformity, if any
- Issue of Test Report for your reference

Step 2 : Filing of the Application Form

- Arrange the Internal Test Report of product as per test report
- Compiling of all required documents
- Remit the application fees direct to BIS in USD*
- Filling-up the Form-V
- Submission of Application Form with all enclosures
- Get the confirmation about scrutiny of documents by BIS Office

Step 3 : Remittance of Audit Fees & Travelling Expenses for Inspection

- Pay the audit fees for inspection
- Get the confirmation about BIS Inspection Team
- Organize the VISA, Ticket, Lodging & Boarding for visiting team
- Inspection team will visit to Factory for Physical Verification
- Team will verify the Factory Set-up & Quality Control Process
- Team will pick the random sample to be tested in BIS Approved Lab

Step 4 : Organize the Lab Test Report (LTR)

- Select the BIS recognized lab as per guideline of BIS Office
- Provide the sample (picked-up by team) of product to get tested
- Get the test result

Step 5 : Closure of Non-Conformity (if any) and Remit the Official Fees

- Rectify the process of Factory Unit, if inspection team points out
- Submit the proof of rectification done by unit to close non-conformity
- Pay the annual marking fees (annual license fees) to BIS
- Submit the Bank Guarantee to BIS

Step 6 : Get the ISI Marking /Registration Certificate

Conformity documents

Documents

What are the conformity documents ? Can you please list them ?

Not Applicable

For ISI Certificate, refer to comments sections of "Certifications/Registrations".

Format

Does it exist a specific format for the conformity documents ? If yes, can you please provide us an example in english ?

No Particular Format

Comments

Comments

Importation rules

General rules

What are the general rules in order to import the product ?

Generally, Importers are required to register with the DGFT to obtain an Importer Exporter Code Number (IEC) issued against their Permanent Account Number (PAN), before engaging in EXIM activities. After an IEC has been obtained, the source of items for import must be identified and declared.

IMPORT PROCEDURE

Step1 : File Bill of Entry with Business Identification Number (BIN)

Step 2: Determine rate of duty for clearance from warehouse

Step 3: File requisite documents with customs department

Step 4: Submit import report/ manifest

Step 5: Receive permission to import goods

Documents

What are the required documents ?

Mandatory Documents:

1. Importer Exporter Code issued from the DGFT
2. Bill of Entry
3. Examination Order generated by the EDI system of Customs

Optional Documents depending upon the type of Product being Imported:

1. Certificate of Origin issued by Authorized Person/ Agency at the place of manufacturing/ processing etc. of the food consignment. Certificate of Origin shall contain information on Country of Origin etc. Hence if the consignor is from a different country;
2. Stuffing list, Packing List;
3. Commercial invoice as mentioned in the Bill of Entry (BoE);
4. Bill of Lading as mentioned in the Bill of Entry (BoE) for sea consignment;
5. Air Way Bill as mentioned in the Bill of Entry (BoE) for air consignment;

Time period

In which time period may we obtain the right to import ?

It generally takes 45 to 60 days for completing the overall procedure from import clearance to Product reaching store shelf.

Administration

What is the administration in charge of importations ? Name and address ?

- Directorate General of Foreign Trade
- Central Board of Excise and Customs

Comments

Comments

Inspection of the product

Inspection method

Does it exist an inspection procedure applicable to the product ? If yes, which administration is in charge of ? Can you please describe the procedure ?

IS 10322-5-5 (1987) : Luminaires, Part 5: Particular requirements, Section 5: Flood Light (Superseding IS:1947) [ETD 24: Illumination Engineering and Luminaries]

i.e Indian Standard – SPECIFICATION FOR LUMINAIRES -Part 5 Particular Requirements-Section 5 Flood-Lights

3. GENERAL TEST REQUIREMENTS

3.1 The provisions of 3 of Part 1 of this standard shall apply. The tests shall be carried out in the order listed in this standard.

13. TESTS

13.1 Classification of Tests

13.1.1 Type Tests – The following shall constitute the type tests:

- a) Visual examination (see 13.2),
- b) Mechanical strength test (see 13.3)
- c) Endurance tests and thermal tests (see 13.4),
- d) Resistance to dust and moisture (see 13.5)
- e) Insulation resistance and electric strength (see 13.6),
- f) Resistance to heat, fire and tracking (see 13.7), and
- g) Photometric test (see 13.8).

13.1.1.1 Criterion of acceptance – Samples for type testing shall be mutually agreed between the purchaser and the supplier. These samples shall be submitted for testing together with relevant data. The testing authority shall issue a type approval certificate if the luminaires comply with the requirements given in 13.1.1.

In a case of failure in any one type test, the testing authority may call for fresh samples not exceeding twice the number of original samples and subject them to test in which failure occurred. If in repeat tests, no failure occurs, the tests shall be considered to have been satisfied.

13.1.2 Acceptance Tests – The following shall constitute the acceptance tests;

- a) Visual examination (see 13.2),
- b) Resistance to dust and moisture (see 13.7),
- c) Insulation resistance and electric strength (see 13.7), and
- d) Photometric test (see 13.8).

13.1.2.1 The number of samples for acceptance shall be as agreed upon between the purchaser and the supplier. However, a recommended plan of sampling is given in Appendix A.

13.1.3 Routine Tests – The following shall constitute the routine tests:

- a) Visual examination (see 13.2), and
- b) Insulation resistance and electric strength (see 13.6).

13.2 Visual Examination – The luminaires shall be examined visually for external finish, workmanship and electrical connections.

13.3 Mechanical Strength Test – The provisions of 5 of Part 4 of this standard shall apply.

13.4 Endurance Test and Thermal Test – The provisions of 6 of Part 4 of this standard shall apply.

13.5 Resistance to Dust and Moisture – The provisions of 2 of Part 4 of this standard shall apply.

13.6 Insulation Resistance and Electric Strength Test – The provisions of 3 of Part 4 of this standard shall apply.

13.7 Resistance to Heat, Fire and Tracking – The provision of 7 of Part 4 of this standard shall apply.

13.8 Photometric Test – The photometric performance shall be determined by a suitable photometric tests as agreed between the purchaser and the supplier. A recommended photometric test for flood-lights fitted with tungsten filament lamps is given in Appendix B.

13.8.1 Flood-lights fitted with tungsten filament lamps shall meet the photometric requirements specified in 12.

13.8.2 The photometric performance shall be determined by the test method given in IS 13383 (Part 3): 1992 Photometry of luminaires – Method of measurement : Part 3 Luminaires for flood lighting.

APPENDIX A (for Clause 13.1.2.1) SAMPLING PLAN FOR ACCEPTANCE TESTS (Please refer to PDF file attached)

APPENDIX B (For Clause 13.8) PHOTOMETRIC TEST FOR FLOOD-LIGHTS (Please refer to PDF file attached)

IS 418:2004 i.e Indian Standard-TUNGSTEN FILAMENT LAMPS FOR DOMESTIC AND SIMILAR GENERAL LIGHTING PURPOSES (Fourth Revision)

12 SAMPLING

12.1 Principles of Sampling

The lamps for testing shall be selected so as to ensure proper representation.

NOTE — It should first be ascertained that the values of the rated luminous flux comply with the requirements of the relevant lamp data sheet.

It is not necessary to replace an accidentally broken lamp if the result of the test (approval or rejection) is not affected by its replacement, provided the required quantity of lamps for any subsequent test is available. If replaced, a broken lamp shall be neglected in calculating the test results.

NOTE — Accidentally broken lamps include, for example, lamps damaged during handling and transportation and also lamps becoming defective for reasons, which are not connected with the purpose of a particular test being applied.

For batch and comparability testing, some lamps additional to the test quantity shall be selected. These lamps shall only be substituted for lamps of the test quantity, if necessary to make up the required number of lamps for the test.

12.2 Sampling for Whole Production Testing (Applicable for Manufacturers)

12.2.1 Pre-compliance Testing for Certification Purposes

Sampling for the pre-compliance test is given in C-2.

12.2.2 Manufacturer's Test Data

The manufacturer shall make available all the data of his finished product tests so far as these relate to the lamp types on the manufacturer's nominated list and are pertinent to the requirements of this standard.

These data shall refer to a sufficient number of lamps, selected over a 12-month period, so as to be representative of the whole production. To meet this requirement there shall be provided:

Manufacturers' test data on:

— for the four largest groups (or all groups if there are less than four), at least 200, 300 and 200 lamps for, respectively, ITQ, RTQ and LTQ with a minimum of 40, 60 and 40 lamps per group for, respectively, ITQ, RTQ and LTQ. If the LTQ of 200 represents more than 0.01 percent of the production, then only 0.01 percent or 40 lamps, whichever is the greater, need be tested;

— for each of the other groups, which together with the four largest groups make up at least 75 percent of the production, a minimum of 20, 30 and 20 lamps for, respectively, ITQ, RTQ and LTQ;

— where a number of types make up a group, test quantities shall be selected from each of those types which make up at least 50 percent of the production for that group;

— for each type for which data has to be presented to meet the above requirements a minimum of 20, 30 and 20 lamps for, respectively, ITQ, RTQ and LTQ;

Providing the above requirements have been met, any types in the manufacturer's records for which there are less than the quantities for ITQ, RTQ and LTQ of 20, 30 and 20, respectively, shall not be considered.

All tests need not necessarily be carried out on the same lamps. The RTQ may contain other individuals than the ITQ, but the LTQ shall be carried out on individuals selected at random from lamps, which have passed the rating test.

NOTE – This is because the life test results are dependent on the rating test results, this not being the case for inspection test results.

As it may be difficult at the time of sampling to forecast the 12-month production of a type, as a fraction of the whole, percentage values in this section are to be regarded as guidelines, and some flexibility is permitted, provided that the manufacturer's selection of test samples is designed to give proper representation and the minimum test quantities are met.

Where a major change in types made by a factory creates a situation where the minimum test quantities are not met in the 12-month period, it shall be sufficient to show that the rate of testing at the time was compatible with the requirements of this clause.

12.3 Sampling for Batch Testing

12.3.1 There shall be selected a random sample for an ITQ consisting of 50 lamps.

12.3.2 There shall be selected at random an RTQ comprising 100 lamps. The ITQ can be used as part of the RTQ.

12.3.3 From lamps, which have passed the rating test. there shall be selected at random an LTQ of 50 lamps.

Commercialization rules

Notification / Declaration

What are the general rules in order to commercialize a product ?
Is there a notification / declaration to do ?

Trade license will have to be acquired which is governed by Municipality Act- which will location specific. Notification is location specific.

Documents

What are the required documents ?

General documents will be required-in relation to establishment of business.

Delay

What is the delay to obtain the right to commercialize ?

Not Applicable

Display in store

Is there some specific rules on the way to display the product in store ?

Not Applicable

Display on internet

Is there some specific mentions to display the product on internet ?

Information to consumers

Do some specific information to have be given to consumers when the product is placed on the market ?

Not Applicable

Comments

Comments

Administration

What is the administration in charge of commercialization ? Name and address ?

Department of Legal Metrology which comes under the Department of Consumer Affairs